Urban Flood Vulnerability Mapping in Pennsylvania Using the HAND Model

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1. Methods

This study estimates potential inundation depth across Pennsylvania from the USGS 1-arc-second (~30 m) National Elevation Dataset. Twenty-six DEM tiles covering the state were downloaded programmatically from the USGS TNM staged products; the tile *n39w074*, which lies entirely over the Atlantic, does not exist in the archive and was skipped automatically. For each tile, the *pysheds* library was used to (i) condition the DEM by filling pits, filling depressions, and resolving flats; (ii) compute D8 flow directions and flow accumulation; and (iii) derive the Height Above Nearest Drainage (HAND) — the vertical distance of every cell above the nearest drainage channel, with channels defined as cells whose accumulation exceeds 200 upstream cells. Assuming a constant channel water depth of 3 m, potential inundation depth was estimated as 3 – HAND wherever $0 \leq \text{HAND} < 3$ m; negative HAND artifacts at ocean and no-data boundaries were masked. The per-tile inundation rasters were then mosaicked into a single statewide GeoTIFF (25,212 × 14,412 cells, NAD83) and rendered as the map in Figure 1.

2. Results

The statewide map shows potential inundation forming dendritic patterns that closely follow the stream network, with the most extensive low-HAND corridors along the Susquehanna, Delaware, and Allegheny river valleys. Under the 3 m channel-depth scenario with channels defined at an accumulation of 200 cells, roughly 27% of the mosaicked footprint carries a non-zero inundation depth. The footprint is rectangular because 1°×1° DEM tiles do not align with the state boundary, so it includes portions of neighboring states; the white rectangle in the southeast is the missing ocean tile.

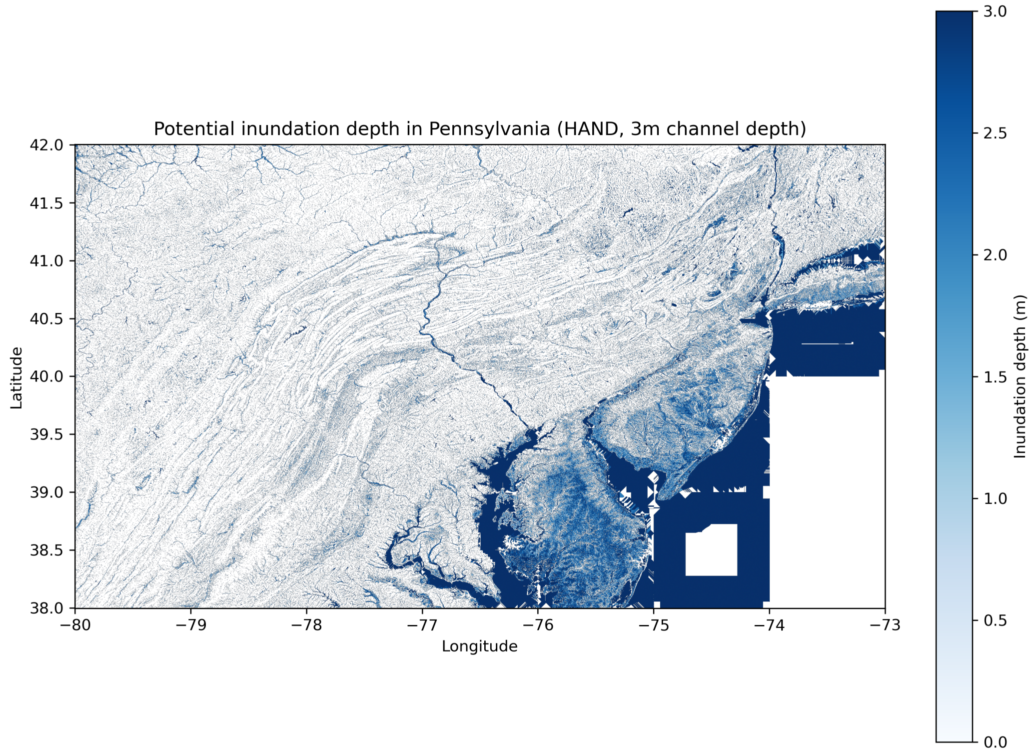

Figure 1. Potential inundation depth in Pennsylvania under a constant 3 m channel depth (HAND model; channels defined as flow accumulation > 200 cells). Blue tones show inundation depth in meters; white areas are outside the modeled flood zone or lack data.

3. Discussion

The two governing parameters are the assumed channel depth (3 m) and the channel-definition threshold (accumulation > 200 cells). Raising the channel depth to 5 m extends the inundation footprint to every cell with HAND < 5 m and visibly enlarges the flooded area. Lowering the accumulation threshold treats even smaller headwater streams as channels, increasing drainage density and drawing more valley floors into the inundated zone; this is why about 27% of the mapped footprint shows potential inundation, and raising the threshold (for example to 1,000 cells) would confine the map to larger rivers.

Several limitations apply. The constant-channel-depth assumption ignores hydraulic routing: real flood depths vary with discharge and channel geometry, so a uniform 3 m likely overstates risk along small streams while understating it on large rivers during severe events. Because each tile was processed independently, flow accumulation stops at tile boundaries, which makes HAND values near tile edges less reliable for rivers that cross them; conditioning a statewide seamless DEM before tiling would reduce this artifact. Ocean and no-data boundaries also produce negative HAND values that had to be masked with a HAND ≥ 0 guard — a data issue that does not arise in inland demonstration areas. Finally, the rectangular mosaic includes neighboring-state areas that clipping to the Pennsylvania boundary would remove.

The full-resolution statewide GeoTIFF (PA_inundation.tif, 25,212 × 14,412 float32 cells, LZW-compressed) and all processing code are available on request.